Inference for two-factor ANOVA model:

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LSM

Recall: Scheffé procedure for simultaneous confidence intervals for arbitrary set of contrasts

Exc. Find confidence band for the regression line $y = \beta_0 + \beta_1 x$ in simple linear regression.

$\mu(x) = \beta_0 + \beta_1 x \qquad Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

$\hat{\mu}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ where $\begin{pmatrix}\hat{\beta}_0\\ \hat{\beta}_1\end{pmatrix} = \hat{\beta}$ is the LSE of $\beta = \begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix}$

$\sigma^2_{\hat{\mu}}(x) = Var(\hat{\mu}(x)) = \sigma^2 \begin{pmatrix}1\\ x\end{pmatrix}^T Var\,\hat{\beta} \begin{pmatrix}1\\ x\end{pmatrix}$

$s^2_{\hat{\mu}}(x)$ " replace by $\hat{\sigma}^2$ (MSE)

Want $K^{1-\alpha} > 0$ s.t.

$$\mathbb{P}\left[\left|\frac{\hat{\mu}(x) - \mu(x)}{s_{\hat{\mu}}(x)}\right| \leq K^{1-\alpha} \text{ for all } x\right] \geq 1-\alpha$$

$$\sup\left|\frac{\hat{\mu}(x) - \mu(x)}{s_{\hat{\mu}}(x)}\right| = \sup\left|\frac{\begin{pmatrix}1\\ x\end{pmatrix}^T(\hat{\beta} - \beta)}{\sqrt{MSE}\sqrt{\begin{pmatrix}1\\ x\end{pmatrix}^T(\mathbb{X}^T\mathbb{X})^{-1}\begin{pmatrix}1\\ x\end{pmatrix}}}\right|$$

$$\leq \underbrace{\frac{1}{\sqrt{MSE}} \sup_v \left|\frac{v^T(\hat{\beta} - \beta)}{\sqrt{v^T(\mathbb{X}^T\mathbb{X})^{-1}v}}\right|}_{\to \sim \sqrt{2F_{2,n-2}}} \quad — *$$

Set $u = (\mathbb{X}^T\mathbb{X})^{-1/2} v$

$$* = \sqrt{\sup_u \frac{\|u^T(\mathbb{X}^T\mathbb{X})^{1/2}(\hat{\beta} - \beta)\|^2}{u^T u}}$$

$$= \lambda_{max}\left((\mathbb{X}^T\mathbb{X})^{1/2}(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T(\mathbb{X}^T\mathbb{X})^{1/2}\right)$$

$$= (\hat\beta-\beta)^T(X^TX)(\hat\beta-\beta)$$
$$\approx \sigma^2\chi^2_2$$

$$\hat\sigma^2 = MSE = \frac{SSE}{n-2}, \quad \hat\beta \perp\!\!\!\perp SSE, \quad \frac{SSE}{\sigma^2}\sim\chi^2_{n-2}$$

<u>Two-way ANOVA</u>.

Individual parameters of interest

$\mu_{\cdot\cdot}, \mu_{i\cdot}, \mu_{\cdot j}, \mu_{ij}$ ← mean for treatment of $(i,j)^{th}$ group

- $\mu_{\cdot\cdot}$: overall mean
- $\mu_{i\cdot}$: mean for i^{th} level of Factor A
- $\mu_{\cdot j}$: mean for j^{th} level of Factor B
- μ_{ij}: mean for $(i,j)^{th}$ treatment group

$\left(\text{Under } \sum\alpha_i=0, \sum\beta_j=0, \sum_i \gamma_{ij}=0\ \forall j, \ \sum_j \gamma_{ij}=0\ \forall i\right)$

In the absence of interaction effect, the two way ANOVA model is said to additive

(Full) In general: $\mu_{ij} = \mu_{\cdot\cdot}+\alpha_i+\beta_j+\gamma_{ij}$

(Reduced) No interaction: $\mu_{ij} = \mu_{\cdot\cdot}+\alpha_i+\beta_j$ (additive)

$$\hat\mu_{ij}^{full} = \overline{Y}_{ij\cdot} \sim N\left(\mu_{ij}, \frac{\sigma^2}{m}\right)$$

$$\hat\mu_{ij}^{red} = \hat\mu_{\cdot\cdot}+\hat\alpha_{i\cdot}+\hat\beta_{\cdot j} = \overline{Y}_{i\cdot\cdot}+\overline{Y}_{\cdot j\cdot}-\overline{Y}_{\cdot\cdot\cdot}$$

$$Var\left(\hat\mu_{ij}^{red}\right) = \frac{\sigma^2}{m}\left(\frac{1}{b}+\frac{1}{a}-\frac{1}{ab}\right) < \frac{\sigma^2}{m} \quad \left[\because \left(1-\frac{1}{a}\right)\left(1-\frac{1}{b}\right)>0\right]$$
$$\forall\ a,b\geq 2$$

This means, for $(1-\eta)$ for C.I. for μ_{ij} under the no interaction model, the expression is

$$\left(\overline{Y}_{i\cdot\cdot}+\overline{Y}_{\cdot j\cdot}-\overline{Y}_{\cdot\cdot\cdot}\right) \pm t^{1-\eta}_{(m-1)ab}\sqrt{\frac{MSE}{m}}\sqrt{\frac{1}{a}+\frac{1}{b}-\frac{1}{ab}}$$

Scheffe's procedure for finding simultaneous confidence intervals for contrasts of treatment group means i.e. for parameter of the form $L=\sum_i\sum_j c_{ij}\mu_{ij}$ where $\sum_i\sum_j c_{ij}=0$

<u>Full model:</u>

Est. $$\hat{L}=\sum\sum c_{ij}\hat{\mu}_{ij}$$
$$=\sum\sum c_{ij}\overline{Y}_{ij}$$
$$\mathrm{Var}\,\hat{L}=\sum\sum c_{ij}^2\,\mathrm{Var}\,\overline{Y}_{ij}$$
$$=\frac{\sigma^2}{m}\sum\sum c_{ij}^2$$

Scheffe's Multiplier is $\sqrt{(ab-1)F^{1-\eta}_{ab-1,\,\underbrace{(m-1)ab}_{df(SSE)}}}$

ab = # treatment groups.

Compare with the problem of finding simultaneous C.I. for contrast involving factor level means corresponding to factor A (i.e., $L=\sum_{i=1}^{a}c_i\mu_{i\cdot}$ where $\sum c_i=0$)

<u>Note:</u> $$\sum_{i=1}^{a}c_i\mu_i=\sum_{i=1}^{a}c_i\frac{1}{b}\sum_{j=1}^{b}\mu_{ij}=\sum_{i=1}^{a}\sum_{j=1}^{b}\frac{c_i}{b}\mu_{ij}$$

If we use the Scheffe procedure for treatment mean	If we use the Scheffe procedure tuned to the factor A level mean
$L = \sum_i \sum_j c_{ij}\mu_{ij}$	$L = \sum_i \sum_j c_{ij}\mu_{ij}$
$\hat{L} = \sum_i \sum_j c_{ij}\overline{Y_{ij\cdot}}$	$\hat{L} = \sum_i \sum_j c_{ij}\overline{Y_{ij\cdot}}$
$s^2(\hat{L}) = \frac{MSE}{m}\sum c_{ij}^2$	$s^2(\hat{L}) = \frac{MSE}{bm}\sum c_{ij}^2$
$S^{1-\eta} = \sqrt{(ab-1)F^{1-\eta}_{ab-1,(m-1)ab}}$	$S^{1-\eta} = \sqrt{(a-1)F^{1-\eta}_{a-1,(m-1)ab}}$

Question: Suppose we can drop the interaction term

i.e., work with the reduced model $\mu_{ij} = \mu_{\cdot\cdot} + \alpha_i + \beta_j$

We want simultaneous C.I. for parameter

$L = \sum\sum c_{ij}\mu_{ij}$ where $\sum\sum c_{ij} = 0$

$$\hat{L} = \sum\sum c_{ij}\hat{\mu}_{ij}^{red} = \sum\sum c_{ij}(\overline{Y_{i\cdot\cdot}} + \overline{Y_{\cdot j\cdot}} - \overline{Y_{\cdot\cdot\cdot}})$$

$$= \sum\sum c_{ij}(\overline{Y_{i\cdot\cdot}} + \overline{Y_{\cdot j\cdot}}) = \sum\sum \tilde{c}_{ij}\overline{Y_{ij\cdot}}$$

Note: $\sum_i\sum_j c_{ij}(\overline{Y_{i\cdot\cdot}} + \overline{Y_{\cdot j\cdot}})$

$$= \sum_i A_i\overline{Y_{i\cdot\cdot}} + \sum_j B_j\overline{Y_{\cdot j\cdot}}$$

est of $\sum A_i\mu_{i\cdot}$

contrast

est of $\sum B_j\mu_{\cdot j}$

where $A_i = \sum_j c_{ij}$ $\quad B_j = \sum_i c_{ij}$

$\sum_i A_i = \sum_i\sum_j c_{ij} = 0$ $\qquad \sum_j B_j = \sum_i\sum_j c_{ij} = 0$